



Important Formulas: The Other Side of Zero

1. Positive Numbers are numbers with a '+' sign in front, such as +1, +2, and +3. They represent numbers greater than zero.

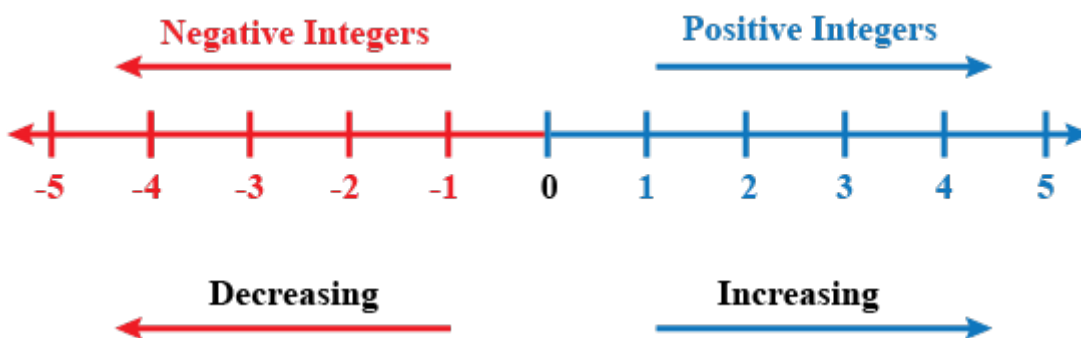
2. Negative Numbers are numbers with a '-' sign in front, such as -1, -2, and -3. They represent numbers less than zero.

3. Zero (0) is a number that comes before 1 and is neither positive nor negative. It represents nothing.

4. Inverse of a Number: The inverse of a number means its opposite sign. For example: Inverse of +3 is -3, and the inverse of -3 is +3.

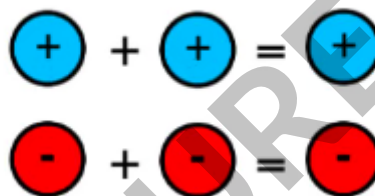
5. Integers:

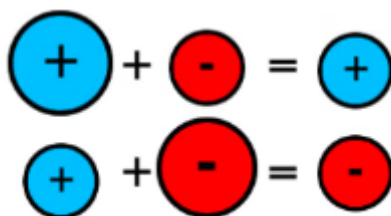
- Positive and negative whole numbers, including zero.
- Example: ..., -3, -2, -1, 0, 1, 2, 3, ...



6. Adding Integers:

- $(+a) + (+b) = (+a + b)$: Adding two positive numbers results in a positive number.
- $(-a) + (-b) = -(a + b)$: Adding two negative numbers results in a more negative number.





7. Subtracting Integers:

- **Subtracting a positive number:**

$$a - b = a + (-b)$$

- **Subtracting a negative number:**

$$a - (-b) = a + b$$

- **Subtracting a larger number from a smaller one:**

Result is negative.

$$\text{Example: } 3 - 7 = -(7 - 3) = -4$$

Integer's Sign	Operation	Sign After Operation
$(+a) - (+b)$	Subtract	+
$(+a) - (-b)$	Add	+
$(-a) - (+b)$	Add	-
$(-a) - (-b)$	Subtract	-

8. Comparing Numbers:

- Positive numbers are greater than negative numbers.
- Numbers increase as you move to the right on the number line and decrease as you move to the left.

9. Number Line:

- A representation of numbers where positive numbers are on the right side of 0, and negative numbers are on the left side.
- Movement to the right represents addition, and movement to the left represents

subtraction.

10. Key Operations on the Number Line:

- **Starting Number + Movement = Target Number**

Example: $5+4=9$

- **Target Number - Starting Number = Movement Needed**

Example: $9 - 5 = 4$

11. Unmarked Number Line (UNL) for Addition and Subtraction:

- The unmarked number line shows the position of zero, but other numbers are not marked.
- You can visualize the addition and subtraction of integers on this line by imagining the scale and the positions of numbers.

12. Addition Example on UNL:

Example: $85+(-60)$

- Visualize starting at 85 and moving 60 steps backward.
- $85 + (-60) = 25$

13. Subtraction Example on UNL:

Example: $-100 - (+250)$

- Visualize starting at -100 and moving backward by 250.
- $-100 - 250 = -350$.

14. Converting Subtraction to Addition:

Subtraction can be converted to addition by replacing the number being subtracted with its inverse.

- Example: $(+7) - (+5) = (+7) + (-5)$
- Similarly, addition can be converted to subtraction by replacing a number with its inverse.

15. Credits and Debits in Banking:

- **Credit:** A positive number representing a deposit into the account.

- **Debit:** A negative number representing a withdrawal from the account.
- Total bank balance = sum of all credits (positive) and debits (negative).
- Example: Start with ₹100, add a ₹60 credit, then subtract ₹30 debit. The balance is ₹130.

16. Geographical Cross-Sections:

Heights are measured from sea level.

- Heights above sea level are represented as positive numbers.
- Heights below sea level are represented as negative numbers.